

BACKTRACK: final judge review

Verdict

The GPS idea and visible removal of review can earn attention. The current product is usable enough to support a serious pilot proposal. It is not yet evidence of learning impact, and I would not rank it first without seeing the field validation and competing entries.

The most persuasive next evidence is an educator saying which current lesson this could serve, a learner showing where the route helps or confuses them, and a school agreeing to a workable routine. Another feature cannot substitute for those facts.

Attention test

Can a judge retell it? Yes: a learner is stuck on today's lesson; BACKTRACK finds a useful earlier step, sends the learner into Khan practice, and checks the return. The GPS metaphor explains this quickly.

Does the site give more than the deck? Yes: the route responds to answers, practice choices, and fresh checks. The deck should stay open as a static fallback rather than spending the entire five-minute slot navigating screens.

What is memorable? A missed turn changes the route while the goal stays. Demonstrated knowledge removes real review. The final task returns to the reason the learner started.

What is still ordinary? A topic selector, progress record, embedded video, and confidence controls are familiar features. None should be sold as the innovation by itself. The complete school recovery workflow needs to earn its advantage through use.

Readability and usability changes made

- Replaced the rejected dark olive/lime styling with a predominantly white interface based on the actual Khan homepage.
- Used one interface font and consistent mathematical notation. Removed the old squiggle mark and overly compressed headline.
- Centered the homepage on the GPS idea and one interaction. Removed the long competition narrative from the main product entry.
- Shortened the learning flow to a task, a short response, and a next action. Kept explanations behind hints or resource choices.
- Kept the destination visible. On mobile, made the route expandable near the top instead of a tiny desktop graph.

- Changed vague factoring instructions into the specific sum and product required by the problem.
- Increased small route labels and key control text after visual inspection.

Product and evidence changes made

- Two distinct fresh unassisted checks now support a route decision. Confidence and a resource open cannot grant a passed skill.
- Added the zero-product step, preserved alternate valid numerical solutions, and checked Unicode minus input.
- Fixed generated questions that could repeat the same expression under a different identifier.
- Fixed comeback checks so a later visit requires new evidence rather than reusing an earlier comeback.
- Removed an unnecessary untested foundation after the learner demonstrates the dependent skill. Historical evidence is preserved; the omitted foundation is not marked mastered.
- Preserved pause/resume and included a human-support path after repeated difficulty.
- Embedded official Khan videos with attribution and a direct original-lesson fallback. Matched practice remains on Khan. No automatic score synchronization is claimed.

Rubric rehearsal

This is an internal readiness judgment, not an organizer score, benchmarked probability, or result forecast. Use the official application weights.

- Needs assessment: 5/10. The problem is specific, but local interviews and school evidence remain incomplete.
- Business and financials: 6/10. The allocation and cost definitions are explicit. Actual quotes, costs, and a continuation payer are unvalidated.
- Innovation: 6/10. The recovery story is clear, but important functionality overlaps with ALEKS, Khan, and Gemini. The workflow must demonstrate value.
- Khan integration: 7/10. The product has matched material and a credible proposed assignment/reporting loop. Partner reporting access and repeated use still need to be established.
- Scale: 6/10. A destination kit is a concrete replication unit. Independent educator reuse has not been observed.
- Impact: 6/10. The retained-reentry definition is stronger than registration counts. No learning result exists yet.

Weighted application readiness: 61/100. This score is intentionally restrained. The limiting factors are evidence and access, not the number of features.

Technical checks and their limits

Fourteen automated tests cover generated mathematics, answer normalization, evidence thresholds, support flags, deeper prerequisites, session blocks, pause/return behavior, local-state validation, and removal of unnecessary review. The production build passes. Public root and nested routes were checked without authentication; the earlier comparison site remains available.

Desktop and mobile layouts were inspected through the browser, including fresh answers and route context. The page's route tools were tested for valid read/pause actions and invalid input. These checks support software behavior; they do not validate educational effectiveness.

The embedded YouTube player was implemented with the official Khan video IDs and the documented referrer policy. The in-app browser did not give a reliable playback view during inspection. Direct Khan links remain visible. Test the embedded player in the school's actual browser and network before a live pitch or learner session. Do not describe playback as universally verified.

What could still make this lose

1. A rival arrives with genuine school relationships and need evidence while BACKTRACK only has plans.
2. Judges see the route as a new interface over a task a teacher already handles well.
3. Teacher reporting and support cost too much time for the benefit.
4. Learners complete the checks but do not gain durable current-class capability.
5. The team confuses a public product event with an independently documented learning outcome.

The plans explicitly test these risks. The next step is to collect the evidence, not remove the caveats or raise the score.

Final presentation advice

Open with the GPS idea. Show one route change. Explain Khan through the learning flow. Give one outcome: back to today's task, still able to do it later. Spend the remaining time on the school routine, fair comparison, cost, and replication. Finish under five minutes.

Keep this sentence available in Q&A: "We have built the experience. The pilot is how we find out whether the recovery routine deserves a place in a school."